Exercise 3:

a) We have that:

(i) $\mathbb{P}(\bigcup_{r=1}^{n} A_r) = 1$

(ii) $\mathbb{P}(A_r) = p \quad \forall r \in \{1, \dots, n\}$

(iii) $\mathbb{P}(A_r \cap A_s) = q, \quad \forall r \neq s$

(iv) $\mathbb{P}(A_r \cap A_s \cap A_t) = 0, \quad 1 \leq r < s < t \leq n$, i.e., no more than two events occur with probability 1.

By σ-subadditivity:

$$\mathbb{P}(\bigcup_{r=1}^{n} A_r) \leq \sum_{r=1}^{n} \mathbb{P}(A_r)$$

$\begin{pmatrix}\text{By (i)}\\ \text{and (ii)}\end{pmatrix} \Rightarrow 1 \leq \sum_{r=1}^{n} p \Rightarrow 1 \leq np \Rightarrow \boxed{p \geq \frac{1}{n}}$

By (ii) and (iii), the events are exchangeable and:

$$1 = P(\bigcup_{r=1}^{n} A_r) = \sum_{k=1}^{n} (-1)^{k-1} \binom{n}{k} \mathbb{P}(\bigcap_{r=1}^{k} A_r)$$

$$= \binom{n}{1}\mathbb{P}(A_1) - \binom{n}{2}\mathbb{P}(A_1 \cap A_2) + \underbrace{\sum_{k=3}^{n}(-1)^{k-1}\binom{n}{k}\mathbb{P}(\bigcap_{r=1}^{k} A_r)}_{0 \text{ (by (iv))}}$$

$$= \binom{n}{1} p - \binom{n}{2} q = np - \frac{n!}{2!\,(n-2)!} q$$

$$\Rightarrow 1 = np - \frac{n(n-1)}{2} q \Rightarrow \frac{-2(1-np)}{n(n-1)} = q \Rightarrow q = \frac{2(np-1)}{n(n-1)}$$

$\begin{pmatrix}\text{since}\\ p \leq 1\end{pmatrix} \quad q \leq \frac{2(n-1)}{n(n-1)} \Rightarrow \boxed{q \leq \frac{2}{n}} \quad \square$

b) We now have (i), (ii) and (iii) as before:

(iv) $\mathbb{P}(A_r \cap A_s \cap A_t) = x$, $1 \leq r < s < t \leq n$

(v) With probability 1, no more than three events A_r occur

Let B_k be the event that exactly k events A_r occur:

(vi) $\mathbb{P}(B_2) + \mathbb{P}(B_3) = 1/2$

(vii) By (v), $\mathbb{P}(B_1) + \mathbb{P}(B_2) + \mathbb{P}(B_3) = 1$

(viii) By (vi) and (vii), $\mathbb{P}(B_1) = \frac{1}{2}$

(ix) By (vi) $\mathbb{P}(B_2) \leq 1/2$ and $\mathbb{P}(B_3) \leq 1/2$.

Since $B_i \cap B_j = \varnothing$ for $i \neq j$ and that $\dot{\bigcup}_{k=1}^{n} B_k = \bigcup_{r=1}^{n} A_k$. Then:

$$A_r = (A_r \cap B_1) \cup (A_r \cap B_2) \cup (A_r \cap B_3)$$

$$\Rightarrow \mathbb{P}(A_r) = \mathbb{P}(A_r \cap B_1) + \mathbb{P}(A_r \cap B_2) + \mathbb{P}(A_r \cap B_3)$$

$$\Rightarrow \sum_{r=1}^{n} \mathbb{P}(A_r) = \sum_{r=1}^{n} \mathbb{P}(A_r \cap B_1) + \sum_{r=1}^{n} \mathbb{P}(A_r \cap B_2) + \sum_{r=1}^{n} \mathbb{P}(A_r \cap B_3)$$

(Since $\sum_{r=1}^{n} \mathbb{P}(A_r) = np$, and exactly k of the A_r events occur within the B_k event)

$$\Rightarrow np = \mathbb{P}(B_1) + 2\mathbb{P}(B_2) + 3\mathbb{P}(B_3)$$

$$\Rightarrow np = \mathbb{P}(B_1) + 2(\mathbb{P}(B_2) + \mathbb{P}(B_3)) + \mathbb{P}(B_3)$$

(by (vi) and (viii)) $\Rightarrow np = \frac{1}{2} + 2 \cdot \frac{1}{2} + \mathbb{P}(B_3)$

$$\Rightarrow np = \frac{3}{2} + \mathbb{P}(B_3)$$

(since $\mathbb{P}(B_3) \geq 0$) $\Rightarrow np \geq \frac{3}{2} \Rightarrow \boxed{p \geq \frac{3}{2n}}$

There are $\binom{n}{2}$ possible intersections $A_r \cap A_s$ $(r \neq s)$, so that $\sum_{1 \leq r < s \leq n} \mathbb{P}(A_r \cap A_s) = \binom{n}{2} q$. Also, since $B_i \cap B_j = \varnothing$ for $i = j$ and $\dot{\bigcup}_{k=1}^{n} B_k = \bigcup_{r=1}^{n} A_k$, then:

$$\sum_{1 \leq r < s \leq n} \mathbb{P}(A_r \cap A_s) = \sum_{1 \leq r < s \leq n} \mathbb{P}(A_r \cap A_s \cap B_1) + \sum_{1 \leq r < s \leq n} \mathbb{P}(A_r \cap A_s \cap B_2) + \sum_{1 \leq r < s \leq n} \mathbb{P}(A_r \cap A_s \cap B_3)$$

$$\Rightarrow \binom{n}{2} q = 0\,\mathbb{P}(B_1) + \binom{2}{2}\mathbb{P}(B_2) + \binom{3}{2}\mathbb{P}(B_3) = \mathbb{P}(B_2) + 3\mathbb{P}(B_3) = (\mathbb{P}(B_2) + \mathbb{P}(B_3)) + 2\mathbb{P}(B_3)$$

(by (vi)) $\Rightarrow \binom{n}{2} q = \frac{1}{2} + 2\mathbb{P}(B_3)$

(by (ix)) $\Rightarrow \binom{n}{2} q \leq \frac{1}{2} + 2\left(\frac{1}{2}\right) \Rightarrow \frac{n(n-1)}{2} q \leq \frac{3}{2} \Rightarrow q \leq \frac{3}{n(n-1)}$

Since intersections of three element are possible, then $n \geq 3$, which implies that $n-1 \geq 2$ and $n(n-1) \geq 2n$. Therefore

$$q \leq \frac{3}{n(n-1)} \leq \frac{3}{2n} \leq \frac{3+5}{2n} = \frac{4}{n} \Rightarrow \boxed{q \leq \frac{4}{n}}$$

□